

## Data Collection and Preprocessing Phase

|               |                                              |
|---------------|----------------------------------------------|
| Date          | 06 September 2026                            |
| Team ID       |                                              |
| Project Title | India' Agricultural crop production Analysis |
| Maximum Marks | 10 Marks                                     |

### Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

| Section                      | Description                                                                                                                          |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Data Overview                | The dataset contains information about crop production in India, including year, district, crop, season, area, production and yield. |
| Data Cleaning                | Check missing values and duplicate records, remove duplicates, handle missing values, and correct formatting errors.                 |
| Data Transformation          | Filter and sort data by year, state, crop and season. Create useful, calculated columns for analysis.                                |
| Data Type Conversion         | Convert year; area, production and yield into the correct data types for analysis.                                                   |
| Column Splitting and Merging | Split or merge columns only when required.                                                                                           |
| Data Modeling                | Create measures for production and yield.                                                                                            |
| Save Processed Data          | Save the cleaned dataset as PDF form.                                                                                                |